

Lightweight state-of-health estimation of lithium-ion batteries based on statistical feature optimization

Houde Dai^{a,b,c}, Jiaxin Wang^{a,b}, Yiyang Huang^a, Yuan Lai^a, Liqi Zhu^{a,*}

^a Quanzhou Institute of Equipment Manufacturing, Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Jinjiang, Fujian, 362215, PR China

^b School of Mathematics and Physics, Lanzhou Jiaotong University, Lanzhou, 730070, PR China

^c Fujian Key Laboratory of Special Intelligent Equipment Safety Measurement and Control, Fujian Special Equipment Inspection and Research Institute, Fuzhou, Fujian, 350008, PR China

ARTICLE INFO

Keywords:

State-of-health
Lithium-ion battery
Statistical features
Health feature optimization
NGO-Dualkernel-GPR

ABSTRACT

To enhance the model estimation performance and minimize the possibility of overfitting, we propose a statistical optimization strategy of healthy features (HFs) for estimating the state-of-health (SOH) of lithium-ion batteries (LIBs). Firstly, a series of initial HFs were extracted from the voltage, current, temperature, incremental capacity (IC) curves, and differential thermal voltammetry (DTV) curves according to the battery characteristics. Secondly, the six statistical features (i.e., mean, median, lower quartile, range, upper quartile, and standard deviation) of the initial HFs for each charge cycle were calculated. Thirdly, to minimize the impact of redundant features and noise, the optimal HF set was identified by a comparative analysis among different combinations of the six statistical features. Thus, the battery SOH can be estimated using the dual-kernel Gaussian process regression (GPR), which improves the estimation accuracy and generalization ability of the single-kernel GPR. To prevent estimation errors due to manual adjustments, the GPR hyperparameters were optimized via the northern goshawk optimization (NGO) algorithm. Experiments were conducted on the NASA and Oxford datasets, with most estimation errors below 1%. Experimental results demonstrate the lightweight NGO-Dualkernel-GPR model, by incorporating the statistical feature optimization strategy, achieves exceptional SOH estimation performance for different LIBs.

1. Introduction

The widespread application of electric vehicles (EVs) has been sparked by the urgent demand to overcome environmental challenges [1,2]. Lithium-ion batteries (LIBs) are crucial for hybrid and pure EVs due to their high energy density and low discharge rate [3,4]. However, their application has ushered in significant challenges [5,6]. One of the crucial challenges is the capacity degradation experienced by the LIB during repeated charge and discharge cycles [7]. Declining capacity of LIBs restricts the driving range of EVs and their safety performance. Measuring the state of health (SOH) of LIBs is an essential function of a battery management system (BMS) [8]. The U.S. Advanced Battery Council (USABC) recommends a SOH threshold of 80% for the retirement of LIBs packs [9,10]. When the battery SOH declines to this threshold, the LIB can no longer be utilized for EVs [11].

Inaccurate estimation of the moment the battery SOH reaches the

retirement threshold can lead to inevitable damage to the EV [12]. Therefore, accurate and efficient SOH estimation is essential for applications of LIBs.

Recently, many in-depth studies have been conducted on the SOH estimation of LIBs. The battery SOH is evaluated by LIB capacity or internal resistance, whereas SOH estimation is usually based on measurable external parameters, such as voltage, current, and temperature. Existing SOH estimation methods for LIBs can be divided into two major categories, i.e., model-based methods and data-driven methods [13,14]. Model-based methods can be implemented by a variety of models, such as the equivalent circuit model (ECM) [15,16] and the electrochemical model (EM) [17–19].

Yu et al. [20] determined the battery structure and initialized parameters by electrochemical impedance spectrum (EIS). Moreover, the correlations between the model parameters and the characteristic voltage were calculated by the iterative learning algorithm with the fractional-order ECM. The correlations were employed for evaluating

* Corresponding author.

E-mail address: liqizhu@fjirsm.ac.cn (L. Zhu).

<https://doi.org/10.1016/j.renene.2023.119907>

Received 10 August 2023; Received in revised form 24 November 2023; Accepted 26 December 2023

Available online 29 December 2023

0960-1481/© 2023 Elsevier Ltd. All rights reserved.

Abbreviations			
SOH	state of health	CV	constant voltage
GPR	Gaussian process regression	LIBs	lithium-ion batteries
IC	incremental capacity	NGO	northern goshawk optimization
BMS	battery management system	DTV	differential thermal voltammetry
ECM	equivalent circuit model	EVs	electric vehicles
SEI	solid electrolyte interface	EM	electrochemical model
SOC	state of charge	EIS	electrochemical impedance spectrum
AI	artificial intelligence	CNN	convolutional neural network
SVM	support vector machine	NNs	neural networks
HF	health feature	PCCs	Pearson correlation coefficients
HI	health indicators	BP	back-propagation
CC	constant current	MAE	mean absolute error
		RMSE	root mean square error
		EOL	end-of-life

the battery structure, where features with superior evaluation results were selected for subsequent SOH estimation. Zhang et al. [21] proposed a novel EIS-based method for battery SOH estimation. An ECM is first introduced, in which the solid electrolyte interface (SEI) resistance and charge transfer resistance are employed to map their relationship with SOH under variant temperature and state of charge (SOC). Ho et al. [22] proposed a novel estimation scheme based on a fractional-order battery model to simultaneously estimate the internal resistance and capacity fade as SOH indicators. The difficulty of the EM method is that the battery SOH degradation is caused by multiple failure mechanisms and requires dimensionality reduction in subsequent calculations [23]. In general, capacity is adopted as a battery ECM parameter with slow time-varying characteristics. Then an adaptive filtering algorithm for parameter identification and state estimation of the model is utilized to estimate the battery capacity [24]. ECM is more concise in battery SOH estimation than EM. However, the simple model structure can hardly reflect the complex aging mechanism of LIBs, yielding inaccurate battery SOH estimation.

With the rapid development of artificial intelligence (AI), data-driven methods have become a research hotspot in SOH estimation [25]. Unlike model-based methods, data-driven methods do not need to consider the electrochemical reaction and failure mechanism inside LIBs. Data-driven methods for battery SOH estimation comprise convolutional neural network (CNN) [26], support vector machines (SVM) [27] Gaussian filters [28], Bayes model [29], random forest [30], and autoencoder [31]. Compared with other methods, neural networks (NNs) have a more vital generalization ability and can obtain higher estimation accuracy. In NNs-based methods, the battery SOH is estimated by extracting the health features (HFs) of the battery aging process. Ma et al. [34] overcome the limitation of the measurement of battery capacity in real applications, some external characteristic parameters related to voltage, current, and temperature are selected from the charging-discharging process as health indicators (HIs) to describe the aging mechanism of the batteries. Neighborhood component analysis (NCA) is used to eliminate redundant information of HIs with high correlation in order to reduce computational burden. Wen et al. [35] evaluated the SOH of LIBs by extracting the HFs from the incremental capacity (IC) curves during the aging process. Then the Pearson correlation coefficients (PCCs) between each extracted HF and the SOH of LIBs were calculated. Finally, the back-propagation (BP) network was utilized for battery SOH estimation. However, the Pearson correlation coefficient between some HFs and SOH indicates that they do not highly correlate with the SOH. Furthermore, Wen et al. did not further process these HFs, making it impossible to comprehensively characterize the battery aging process and achieve high estimation accuracy. After extracting 6 HFs from EIS, Wang et al. [36] calculated the degree of correlation between these 6 HFs and SOH. The Gaussian process regression (GPR) model was employed to estimate battery SOH, where

the features with low correlation were eliminated. The discharge process divides into different stages according to the curvature of the discharge curve, thereby extracting HFs from different stages of the discharge voltage curve [37]. However, the LIB discharge process depends on its operating conditions, such as the operating temperature, discharge current, and cut-off voltage. Compared to the charging process, the discharging process is uncontrollable and difficult to implement. Son et al. [38] extracted multi-physical features from mechanical and electrochemical reactions. The resistance was calculated by taking the ratio of voltage to time in the voltage curve, where the reciprocal of the resistance obtained from the ratio of voltage to current in the current curve is determined as the HFs to estimate the battery SOH. The effectiveness of the estimated SOH method based on the extracted HIs was validated using GPR, SVM, LSTM, and other models. Lin et al. [39] employed the genetic algorithm (GA) to optimize the HFs extracted from voltage and IC curves. Furthermore, a dual-kernel function GPR model was introduced for SOH estimation, where the mean absolute error (MAE) and root mean square error (RMSE) were within 1%. However, the hyperparameters of the dual-kernel GPR model, i.e., the estimation model are calculated manually. The manual calculation cannot guarantee the hyperparameters reach the optimal values and the errors caused by human operations are avoided.

To address the above limitations, as shown in Table 1, this study proposes a battery SOH estimation approach that is optimized by multiple statistical features with the following contributions:

- (1) To reduce the impact of redundant features and noise, an optimal HF set for SOH estimation is acquired by a comparative analysis among different combinations of the 6 statistical features (mean, median, lower quartile, extreme deviation, upper quartile, and standard deviation) calculated from all initial charging feature sets.
- (2) A Dualkernel GPR model is established to significantly improve the estimation accuracy and generalization ability of the GPR with a single-kernel, where an NGO algorithm is introduced to find the optimal hyperparameters of the Dualkernel GPR model.

2. Analysis of health indicators

2.1. Data set description

2.1.1. NASA dataset

The experimental dataset was obtained from the NASA Ames Prediction Center of Excellence, where the three LIBs (labeled No. 5, 6, and 7) were the second-generation 18,650 cells produced at Idaho National Laboratory [40]. All cells were repeatedly operated at room temperature of 24 °C through three different operating modes, i.e., constant current and constant voltage (CC-CV) charge mode, CC discharge, and

Table 1
Summary of the state-of-the-art.

References	Primary method	Limitations
Yu et al. (2020) [20]	Fractional-order modeling of lithium-ion batteries using additive noise assisted modeling and correlative information criterion	A pre-tested nondestructive 1/f noise is difficult to obtain accurate values.
Zhang et al. (2022) [21]	A novel EIS-based method is proposed for battery SOH estimation	Need SEI resistance and charge transfer resistance to map their relationship with SOH under variant temperature and SOC. Difficult to eliminate redundant information the HIs with high correlation.
Ma et al. (2022) [34]	A novel SOH estimation method based on improved long short-term memory (LSTM) and health indicators (HIs) extraction from charging-discharging process is proposed	
Wang et al. (2023) [36]	Extracted health indicators from EIS by DRT method for battery SOH prediction	Difficult to eliminate redundant information the HIs with high correlation.
Son et al. (2022) [38]	Integrated framework for SOH estimation of lithium-ion batteries using multi physics features	Need to consider the high sensitivity of these HIs.
Lin et al. (2023) [39]	A new SOH estimation method for lithium-ion batteries with multi-feature optimization is proposed	Needs constructed the objective function.

impedance measurement mode. The CC charging current, upper voltage limit, and CV charging voltage were 1.5 A, 4.2 V, and 4.2 V, respectively, where the CV ended when the current dropped to 50 mA. CC discharge and CV charge are repeated continuously. The experiment was terminated when the LIB reached the end-of-life (EOL) criterion.

2.1.2. Oxford dataset

Eight KoKam LIBs (Cells 1–8), each cell with a nominal capacity of 740 mAh was tested in a hot chamber set at 40 °C for the Oxford battery degradation dataset [41]. These cells consisted of graphite as the negative material and LiMO₂ as the positive material. LIB capacity was measured every 100 aging cycles to obtain the true SOH. All cells were first charged with CC charging mode until the voltage reached 4.2 V. Each cell was discharged at 740 mA until the voltage was 2.7 V, where current, surface temperature, and capacity values were recorded. Three cells (Cells 1–3) were selected from these eight LIBs for subsequent experiments.

2.2. Feature extraction

Currently, there is no uniform expression for battery SOH [42]. This study defines SOH in terms of capacity attenuation [9], as:

$$SOH = \frac{C_i}{C_o} \times 100\% \quad (1)$$

where C_i indicates the maximum battery capacity when the battery is fully charged in the i th charging/discharging cycle, and C_o indicates the rated capacity (see Fig. 1).

2.2.1. Voltage feature extraction

The charging process of a LIB includes two processes, i.e., CC charging and CV charging. The LIB was first charged in CC mode, which switched to CV charging mode when the voltage reaches 4.2 V. The current gradually decreased until the LIB was fully charged. As shown in Fig. 2(a) and (b), the voltage inflection points of the two LIBs with different operating conditions were located in the 3.85–4.05 V and 3.55–3.75 V intervals, respectively. Prior to this inflection point, the slope of the curve was more significant, while the difference in the value of time at the same voltage was smaller. In contrast, the slope of the voltage curve starts to become smaller after this inflection point, while the time difference at the same voltage gradually becomes more considerable. Considering the characteristics of battery terminal voltage sampling and the calculation complexity in actual operation, the time difference value corresponding to the voltage at a fixed point was selected as the initial voltage feature set input in this study. Selecting a voltage within a specified voltage range can reduce the impact of voltage fluctuations on the feature set. Therefore, the voltage interval in this study was defined as 50 mV.

2.2.1.1. NASA voltage feature. The NASA dataset needs to focus on the voltage variation in the 3.85–4.05 V intervals when extracting features on the voltage. As shown in Fig. 2(a), the voltage fluctuates between 3.8 and 3.9 V, where the time difference corresponding to the same voltage was slight. While observing the inflection point around 3.9 V, the feature range was determined from 3.85 to 4.05 V. The time difference calculated for every 50 mV between 3.85 and 4.05 V was the initial feature set. The initial features extracted in the voltage profile were noted as health indicators (HI_1, HI_2, HI_3, HI_4).

2.2.1.2. Oxford voltage feature. For extracting features on the voltage, the Oxford dataset needs to focus on the voltage variation in the interval between 3.55 and 3.75 V. As shown in Fig. 2(b), the voltage fluctuates between 3.4 and 3.6 V, where the time difference corresponding to the same voltage was small. While observing the inflection point around 3.65 V, the feature range was determined between 3.55 and 3.75 V. The time difference calculated for every 50 mV between 3.55 V and 3.75 V was the initial feature set for SOH estimation. The initial features extracted in the voltage profile were noted as health indicators (HI_1, HI_2, HI_3, HI_4).

2.2.2. Charging current curve

The EOL of a LIB was affected by the positive and negative active

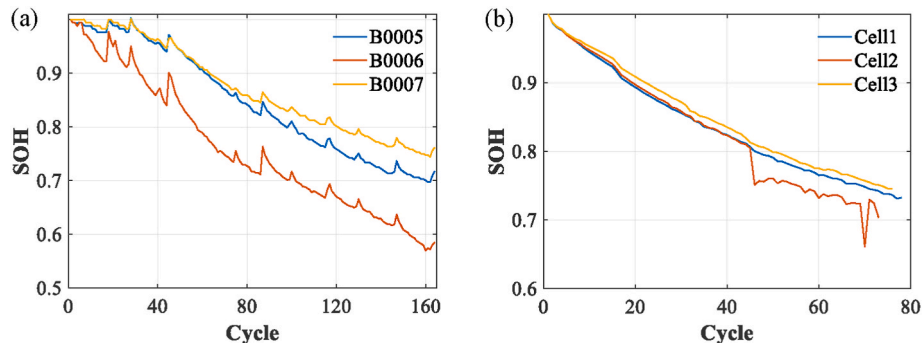


Fig. 1. SOH variation under different cycles. (a) NASA dataset, (b) Oxford dataset.

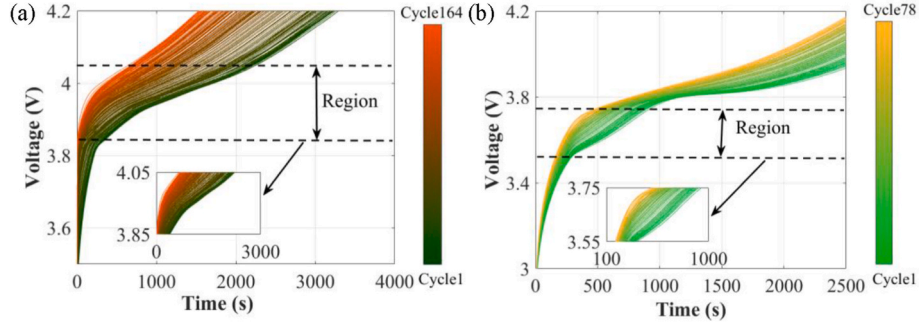


Fig. 2. Charge voltage curves of (a) B0005 from the NASA Dataset and (b) Cell1 from the Oxford dataset.

substances. Charging current is an essential factor that affects the EOL of the LIB. When the internal current of the LIB is too high, resistive heating occurs inside the battery and the active material reacts violently. Therefore, the performance of the LIBs was affected.

Due to the absence of current data in the Oxford dataset, the current feature was only extracted from the NASA dataset. The battery charging current curve was demonstrated in Fig. 3. Besides, the current curves of CC and CV charging modes under different cycles were selected for the battery SOH estimation in this study. Obviously, the overall time of the battery full charging process was approximated with the increase of charge/discharge cycle, where CC time gradually becomes shorter and CV charging time gradually increases. In different charge/discharge cycles, the current in the CV charging mode decreases from 1.5 to 0.2 A, where the current decline time varies accordingly with the battery aging process. Accordingly, we chose the same current difference time interval as the characteristics in the CV charging mode. We set the currents from 1.5 to 0.3 A as the total interval and the time difference of two currents at each interval of 0.3 A as the feature for subsequent SOH estimation. Therefore, we extracted the time differences of current variation ranges (0.3–0.6 A), (0.6–0.9 A), (0.9–1.2 A), and (1.2–1.5 A) as the current curve HIs (HI_5 , HI_6 , HI_7 , HI_8) for SOH estimation, respectively. As Fig. 2 (b) shows, the battery CC charging time gradually decreases as the number of charge/discharge cycles increases. Thus, the battery CC charging time was determined as health factor HI_9 for SOH estimation.

2.2.3. IC curve

The IC curve depicts the ratio of capacity change to voltage variation during the charging and discharging process [23]. Recently, IC curves have been widely utilized to describe the battery aging process. With IC curve analysis, the smooth voltage curve can be converted into a clearly visible peak, which can more accurately reflect the chemical reactions

and electrochemical properties of LIBs. Therefore, IC analysis was well suited for estimating battery SOH. In CC mode, the IC is calculated as:

$$Q = It \quad (2)$$

$$IC = \frac{dQ}{dV} = I \cdot \frac{dt}{dV} \quad (3)$$

where Q is the LIB capacity, t is the charging time, I is the charging current, and V is the voltage.

2.2.3.1. NASA IC Curve. As shown in Fig. 4(b), the two peaks of the IC curve are between the voltage range between 3.8 and 4.1 V. However, the first peak gradually disappears as the LIB ages, where the second peak gradually decreases. Therefore, this study employs the second peak of the IC curve as a characteristic for estimating SOH. The IC curve changes most significantly at 3.8–4.1 V. In this interval, the IC value decreases gradually at the same voltage as the LIB ages. However, in the voltage range near 3.8 V and 4.1 V, the difference in the magnitude of IC values for different cycles at the same voltage was not as pronounced as near the second peak. Considering these two sections because the IC value gap was not apparent, the IC value corresponding to each 50 mV voltage interval in the voltage range at 3.8–4.1 V was selected as the input of the remaining IC curve feature set in this study. The initial health features extracted in the IC curve are marked as $-HI_{10}$, HI_{11} , HI_{12} , HI_{13} , HI_{14} , HI_{15} , HI_{16} , HI_{17} .

2.2.3.2. Oxford IC curve. As shown in Fig. 5(b), the two peaks of the IC curve were between the voltage range from 3.4 to 4 V. However, as the LIB aging, the first and second peaks gradually disappear and decrease, respectively. Therefore, the second peak IC value was adopted as a characteristic for estimating SOH. The values near the inflection point of the IC curve gradually decrease with the number of cycles. Therefore, the IC values corresponding to the voltage interval (3.75–3.9 V), at 50 mV intervals, are selected as the IC-based HIs, i.e., HI_5 , HI_6 , HI_7 , HI_8 , HI_9 .

2.2.4. Temperature profile

The temperature has a significant effect on LIB performance and capacity. As shown in Figs. 6(a) and 7(a), the voltage corresponding to the lowest operating temperature of the LIB gradually increases during the battery aging process. Therefore, the LIB voltage corresponding to the lowest temperature was utilized as a feature input in the temperature profile for estimating battery SOH.

The Differential processing method can improve the SOH estimation accuracy by considering the possible errors and instabilities of temperature sensors. In this study, the temperature variation DT was calculated by approximation to extract the temperature feature set, as:

$$DT(i) = \frac{T(i + \Delta) - T(i)}{\Delta} \quad (4)$$

where T and Δ are the temperature and the sampling interval,

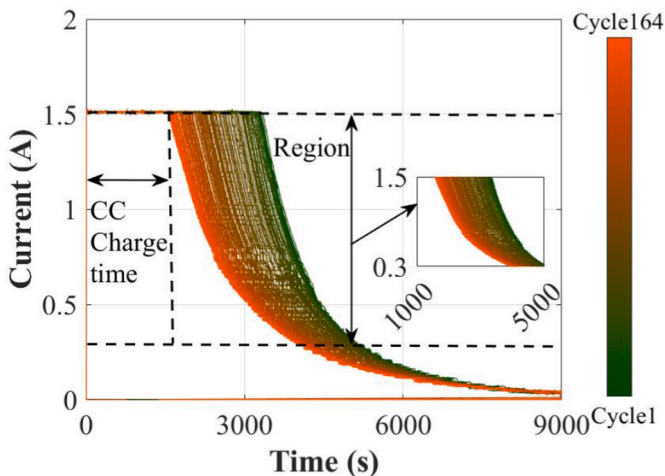


Fig. 3. Charging current curve of B0005 from the NASA Dataset.

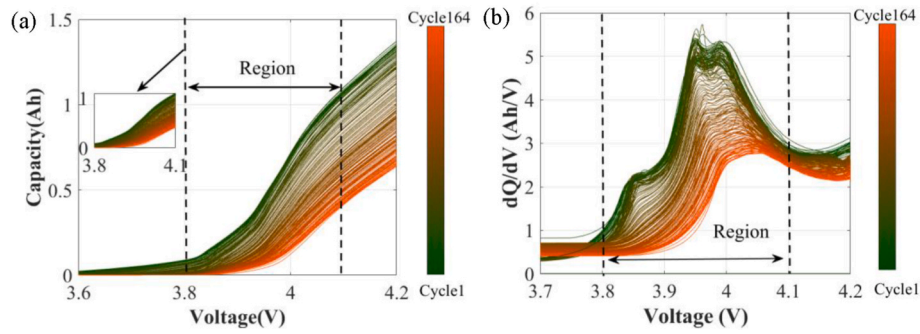


Fig. 4. (a) Capacity curve and (b) capacity increment curve of B0005 from the NASA dataset.

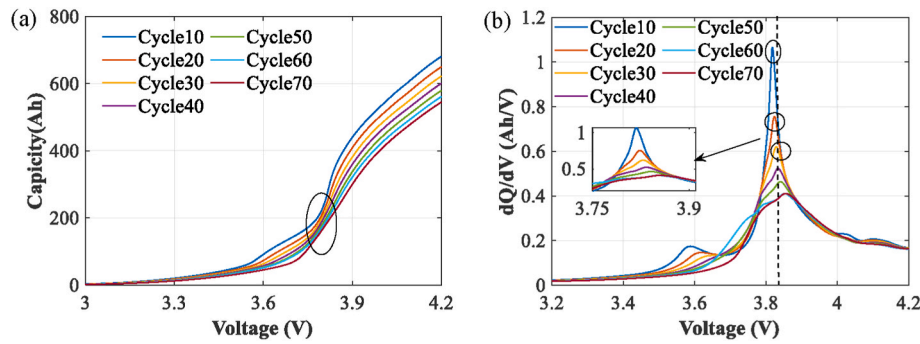


Fig. 5. (a) Capacity curve and (b) capacity increment curve of Cell1 from the Oxford datasheet.

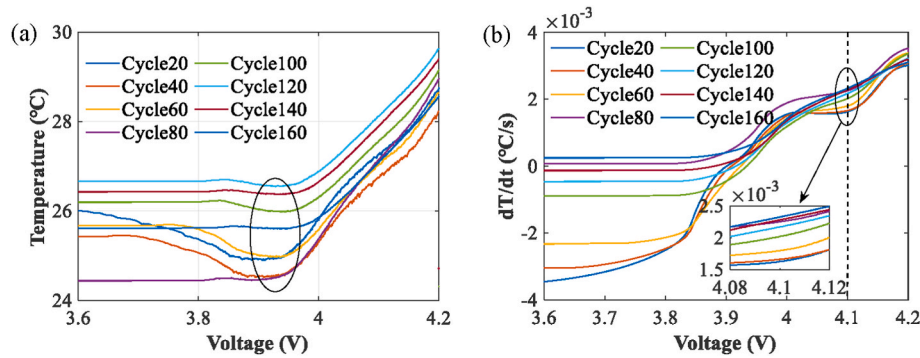


Fig. 6. (a) Temperature curve and (b) DTV curve of B0005.

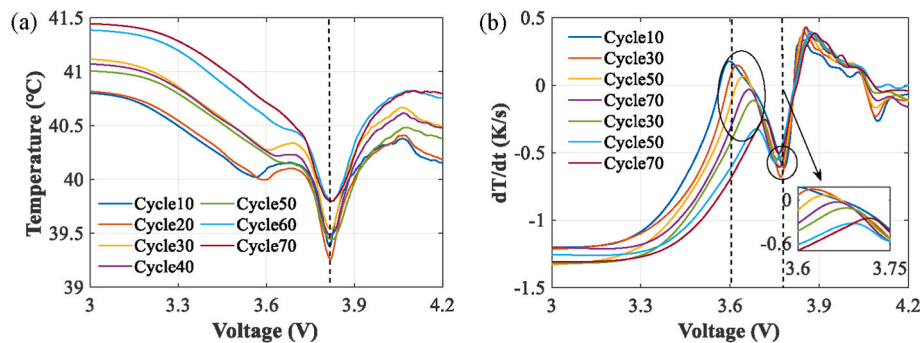


Fig. 7. (a) Temperature curve and (b) DTV curve of Cell1.

respectively.

It may not be able to capture the short-term temperature variation if Δ was too large. Conversely, too small Δ yields insignificant temperature

changes and more significant errors due to the noise. Considering the number of samples per cycle, the sampling interval was 30 mV in this study. Furthermore, the Gaussian process was adapted to smooth the

temperature curve and reduce the influence of noise.

2.2.4.1. NASA temperature profile. As shown in Fig. 6(b), the DTV curve is between 4.08 and 4.12 V, where DT gradually increases with the LIB degradation at the same voltage. Considering the characteristics of LIB terminal voltage sampling and the complexity of calculation in actual operation, this study selected the fixed voltages (4.08 V, 4.1 V, 4.12 V) of the DTV curve, where their corresponding DT values were adopted as the health feature sets. The initial features extracted at the temperature and DT curves are noted as $HI_{18}, HI_{19}, HI_{20}, HI_{21}$.

2.2.4.2. Oxford temperature profile. As shown in Fig. 7(b), the temperature variation of the LIB first decreases and then increases before finally starting to change drastically. With the increase in the number of charge/discharge cycles, the peak value of the LIB gradually decreases and the first valley of the LIB starts to increase. Therefore, we consider the leftmost peak value and the first valley value as feature inputs for estimating battery SOH. Additionally, when the voltage is between 3.6 and 3.75 V, the DT curve decreases along with the increase of the LIB cycles. Hence, the DT value corresponding to the 50 mV interval between 3.6 and 3.75 V was defined as the health feature for the subsequent SOH estimation. The HIs extracted from the temperature profile and DT curves are defined as $HI_{10}, HI_{11}, HI_{12}, HI_{13}, HI_{14}, HI_{15}$.

2.3. Statistical feature set

The battery SOH estimation focuses on the relative variations of initial features. To achieve efficient dimensionality reduction, the different initial features can be analyzed uniformly by statistical methods. Six statistical features (i.e., mean, median, lower quartile, range, upper quartile, and standard deviation) according to statistical distributions of each cycle of all initial feature sets were calculated.

The mean value of each cycle of all the initial feature sets is calculated as:

$$F_A = \frac{\sum_{i=1}^j HI_i}{j} \quad (5)$$

The mean value (F_A) can effectively reflect the trend of a set of data concentrations.

The median of each cycle of all initial feature sets is calculated as:

$$F_M = \begin{cases} \frac{HI_{(j+1)}}{2} & j \text{ is an odd number} \\ \frac{HI_{(\frac{j}{2})} + HI_{(\frac{j}{2}+1)}}{2} & j \text{ is an even number} \end{cases} \quad (6)$$

The median (F_M) can indicate the general level of a set of data.

The lower quartile can well represent the dispersion of a data set. The lower quartile (F_{LQ}) of each cycle for all initial feature sets is calculated as:

$$F_{LQ} = HI_{\left(\frac{j+1}{4}\right)} \quad (7)$$

The extreme difference can well represent the magnitude of a data set. The polar deviation (F_R) of each cycle for all initial feature sets is calculated as follows:

$$F_R = HI_j - HI_1 \quad (8)$$

The upper quartile score can also well represent the dispersion of a data set, whereas it is a value where the data sets are at the upper 75th percentile. The upper quartile (F_{UQ}) of each cycle of all feature sets is computed as:

$$F_{UQ} = HI_{\left(\frac{3(j+1)}{4}\right)} \quad (9)$$

The standard deviation indicates the degree of fluctuation within a data set. The standard deviation (F_{SD}) of each cycle for all initial feature sets is calculated as:

$$F_{SD} = \sqrt{\frac{\sum_{i=1}^j \left(HI_i - \frac{\sum_{i=1}^j HI_i}{j} \right)^2}{j-1}} \quad (10)$$

where HI_i denotes the i th feature, and j represents the number of initial feature sets per cell. j is 21 and 15 in the NASA and Oxford dataset, respectively.

Based on the calculated six statistical features, the statistical feature set ($F_A, F_M, F_{LQ}, F_R, F_{UQ}, F_{SD}$) is formed to estimate battery SOH.

To prove the effectiveness of the feature set extracted in this study, the PCCs (r^2) between each health feature and the battery SOH are shown in Table 2. The PCCs are above 0.95 ($r^2 \leq 1$), indicating that the statistical feature set extracted in this study has high-quality estimation ability [43].

3. Methodology and statistical feature optimization

3.1. NGO algorithm

The northern goshawk (NG) is a medium-large raptor in the hawk family [44]. The NGO algorithm has the characteristics such as fast convergence, high solution accuracy, few algorithm parameters, simple implementation, and multi-objective optimization. In contrast, the particle swarm algorithm (PSO) algorithm has a better global search capability, whereas it is prone to local optima. Although having better diversity and local search capability, the GA algorithm requires more algorithm parameters for adjustment. Although the ability to identify the global optima in large numbers of local optima, the simulated annealing (SA) algorithm only applies to specific problems.

The NG hunting process is divided into two phases. In the exploration phase, the NG seeks out potential prey through prey identification and attacks. In the exploitation phase, the NG pursues and escapes to obtain the final prey. According to the different hunting phases, the mathematical models of the NGO algorithm are described as follows.

3.1.1. Initialization

In the NGO algorithm, the NG population can be represented by the following matrix:

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_N \end{bmatrix}_{N \times m} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i,1} & \cdots & x_{i,j} & \cdots & x_{i,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{N,1} & \cdots & x_{N,j} & \cdots & x_{N,m} \end{bmatrix}_{N \times m} \quad (11)$$

Table 2

PCCs (r^2) between statistical features and battery SOH.

Cell	r^2					
	F_A	F_M	F_{LQ}	F_R	F_{UQ}	F_{SD}
B0005	0.9975	0.9892	0.9896	0.9968	0.9861	0.9970
B0006	0.9939	0.9620	0.9716	0.9927	0.9793	0.9931
B0007	0.9941	0.9859	0.9906	0.9941	0.9796	0.9942
Cell1	0.9937	0.9633	0.9645	0.9989	0.9934	0.9986
Cell2	0.9926	0.9556	0.9520	0.9959	0.9925	0.9986
Cell3	0.9950	0.9699	0.9531	0.9993	0.9958	0.9958

where X is the population matrix of NG, X_i is the location of the i th NG, $x_{i,j}$ is the i th position of the j th dimension of the NG, N is the number of NGs, m is the dimension of the solution problem.

In the NGO algorithm, the objective function to solve the problem can be expressed as a vector of objective functions, as:

$$M = \begin{bmatrix} M_1 \\ \vdots \\ M_i \\ \vdots \\ M_N \end{bmatrix}_{N \times 1} = \begin{bmatrix} M(X_1) \\ \vdots \\ M(X_i) \\ \vdots \\ M(X_N) \end{bmatrix}_{N \times 1} \quad (12)$$

where M is the objective function vector of the NG population, M_i is the objective function of the i th NG.

3.1.2. Phase 1: prey identification (exploration phase)

In the first hunting stage, the NG randomly selects a prey and then quickly attacks it. This phase performs a global search of the search space to determine the optimal area, as:

$$\begin{cases} P_i = X_k, k = 1, 2, \dots, i-1, i+1, \dots, N \\ x_{i,j}^{new,P_1} = \begin{cases} x_{i,j} + r(p_{i,j} - Hx_{i,j}), M_{p_i} < M_i \\ x_{i,j} + r(x_{i,j} - p_{i,j}), M_{p_i} \geq M_i \end{cases} \\ X_i = \begin{cases} X_i^{new,P_1}, M_i^{new,P_1} < M_i \\ X_i, M_i^{new,P_1} \geq M_i \end{cases} \end{cases} \quad (13)$$

where P_i is the prey position of the i th NG, M_{p_i} is the objective function of the prey position of the i th NG, k is a random integer number in the range between 1 and N , X_i^{new,P_1} is the new position of the i th NG, $x_{i,j}^{new,P_1}$ is the new position of the j th dimension of the i th NG, M_i^{new,P_1} is the objective function of the i th NG after the first stage update, r is a random number between 0 and 1, H is an arbitrary integer of 1 or 2.

3.1.3. Phase 2: chase and escape (exploitation phase)

The prey will try to escape when the NG is attacking the prey. Therefore, the NG must keep chasing the prey during the prey-chasing phase. Due to its fast speed, NG can chase prey in almost any situation and eventually successfully capture it. The simulation of this behavior improves the algorithm's ability to search the space locally. This phase is expressed as:

$$x_{i,j}^{new,P_2} = x_{i,j} + R(2r - 1)x_{i,j} \quad (14)$$

$$R = 0.02 \left(1 - \frac{t}{T}\right) \quad (15)$$

$$X_i = \begin{cases} X_i^{new,P_2}, M_i^{new,P_2} < M_i \\ X_i, M_i^{new,P_2} \geq M_i \end{cases} \quad (16)$$

where t and T are the current and maximum number of iterations, respectively. R is the attack radius, which shrinks as the number of iterations increases. $x_{i,j}^{new,P_2}$ is the value of the solution of the i th hawk at the new location in the j th dimension. F_i^{new,P_2} is the value of the objective function at that stage.

3.2. Improved GPR

GPR is a machine learning (ML) algorithm that solves problems such as high dimensionality, small sample size, and nonlinearity [45,46]. For the GPR issue, the input feature x and output y can be modeled as an implicit function plus independent Gaussian noise, as:

$$y = f(x) + \varepsilon, \varepsilon \sim N(0, \delta^2) \quad (19)$$

where ε is the Gaussian noise.

For any input, the GPR is determined by its mean function $m(x)$, the covariance function $K(x_i, x_j)$, and the corresponding probability distribution function $f(x)$ that obeys the Gaussian distribution:

$$\begin{cases} m(x) = E[f(x)] \\ K(x_i, x_j) = E[(f(x) - m(x))(f(x') - m(x'))] \end{cases} \quad (20)$$

The kernel function $K(x_i, x_j)$ plays a vital role in the GPR model by adaptively adjusting parameters to fit different data sets. After the kernel function measures the similarity between samples, the ML model can fit the data more accurately and then improve the estimation accuracy of GPR. Typically, the mean function was set to 0. The squared exponential covariance function is utilized as the kernel function, as:

$$K(x_i, x_j) = \delta_f^2 \exp\left(-\frac{(x_i - x_j)^2}{2\delta_l^2}\right) \quad (21)$$

where hyperparameters δ_f and δ_l are the signal variance and the length scale, respectively. Usually, the degradation process of battery capacity has a long-term dynamic fluctuating trend.

To achieve better SOH estimation, this study improves the GPR by constructing a dual-kernel function. This function captures the dynamic fluctuations of both the long-term and short-term battery degradation processes by linear operations between two separate kernels. This composite kernel function is shown as:

$$\begin{cases} K(x_i, x_j) = (\text{Ma3} + \text{Ma5}) \\ \text{Ma3} = \delta_{f1}^2 \left(1 + \frac{\sqrt{3(x_i - x_j)}}{\delta_{l1}}\right) \exp\left(-\frac{\sqrt{3(x_i - x_j)}}{\delta_{l1}}\right) \\ \text{Ma5} = \delta_{f2}^2 \left(1 + \frac{\sqrt{5(x_i - x_j)}}{\delta_{l2}} + \frac{5(x_i - x_j)^2}{3\delta_{l2}^2}\right) \exp\left(-\frac{\sqrt{5(x_i - x_j)}}{\delta_{l2}}\right) \end{cases} \quad (22)$$

where the dual-kernel function comprises two Matern kernel functions, namely the Matern3 kernel function (Ma3) and the Matern5 kernel function (Ma5). The Ma5 term and Ma3 term capture a smooth long-term downward trend and a smooth short-term change, respectively. Therefore, this dual-kernel function can effectively track both the long-term downward trend and short-term variation in LIB capacity.

If $m(x)$ is zero and noise is considered, the prior distribution of the observations can be expressed as:

$$y \sim N(0, K(x_i, x_j) + \delta_N^2 I_N) \quad (23)$$

where I_N is an N -dimensional unit matrix, and $\delta_N^2 I_N$ denotes the noise covariance matrix.

The joint probability distribution of the observed and predicted values is expressed as:

$$\begin{bmatrix} y \\ f^* \end{bmatrix} = N\left(0, \begin{bmatrix} K(x, x) + \delta_N^2 I_N & K(x, x^*) \\ k(x^*, x) & k(x^*, x^*) \end{bmatrix}\right) \quad (24)$$

where f^* denotes the estimated value of y , and x^* indicates the test data set.

The GPR prediction can be expressed as:

$$p(f^* | x, y, x^*) \sim N(\bar{f}^*, \text{cov}(f^*)) \quad (25)$$

$$\bar{f}^* = K(x, x^*) (K(x, x) + \delta_N^2 I)^{-1} Y \quad (26)$$

$$\text{cov}(f^*) = K(x^*, x^*) - K(x^*, x) (K(x, x) + \delta_N^2 I)^{-1} K(x, x^*) + \delta_N^2 I \quad (27)$$

where $\bar{f}^*$ is the predicted mean vector, and $\text{cov}(f^*)$ is the predicted variance.

The hyperparameter sets $\Theta_1 = [\delta_f, \delta_l, \delta_N]$ and $\Theta_2 = [\delta_{f1}, \delta_{l1}, \delta_{f2}, \delta_{l2}, \delta_N]$ are unknown and can typically be determined by the maximum likelihood method. However, it does not guarantee whether the value of the obtained hyperparameter set is optimal. To avoid the error caused by the manual computation of hyperparameters during the optimum searching, this study adopted NGO for hyperparameter optimization. This proposed algorithm can automatically search for the optimal set of hyperparameters, thereby improving the accuracy of modeling prediction.

4. Experimental results and discussion

The flowchart of the proposed SOH estimation method is presented in Fig. 8. To validate the effectiveness of the proposed method, the comparative experiment was conducted in two parts. The first part aimed to verify the efficacy of the NGO- Dualkernel-GPR models on the NASA and Oxford datasets. The second part was an experiment on the statistical feature sets after optimization. The leave-one-out cross-validation method was adopted in this study, where one LIB in the utilized dataset was employed as the testing set. The remaining batteries on this dataset were employed as the training set. The effectiveness of the improved GPR model was evaluated using MAE and RMSE, as:

$$MAE = \frac{1}{n} |\hat{y}_i - y_i| \quad (28)$$

$$RMSE = \sqrt{\frac{1}{n} (\hat{y}_i - y_i)^2} \quad (29)$$

where $\hat{y}_i$ denotes the predicted value, y_i denotes the true value, and n denotes the sample size of the test datasets.

4.1. NGO-DualKernel-GPR model for SOH estimation

4.1.1. NASA dataset testing

The effectiveness of the NGO-Dualkernel-GPR model was verified in this study. A performance comparison of the following models was conducted in terms of SOH estimation:

(1) the GPR model with all statistical features as inputs and squared exponential covariance function as the kernel, (2) the dual-kernel GPR model with Ma3 and Ma5 kernels, (3) the NGO-optimized dual-kernel GPR model. The battery SOH estimation results of the above models are shown in Fig. 5, where the MAE and RMSE are shown in Table 3.

As shown in Fig. 9, the dual-kernel GPR can estimate the battery SOH more effectively than the single-kernel GPR. However, due to the manual calculation of hyperparameters cannot be optimized, the

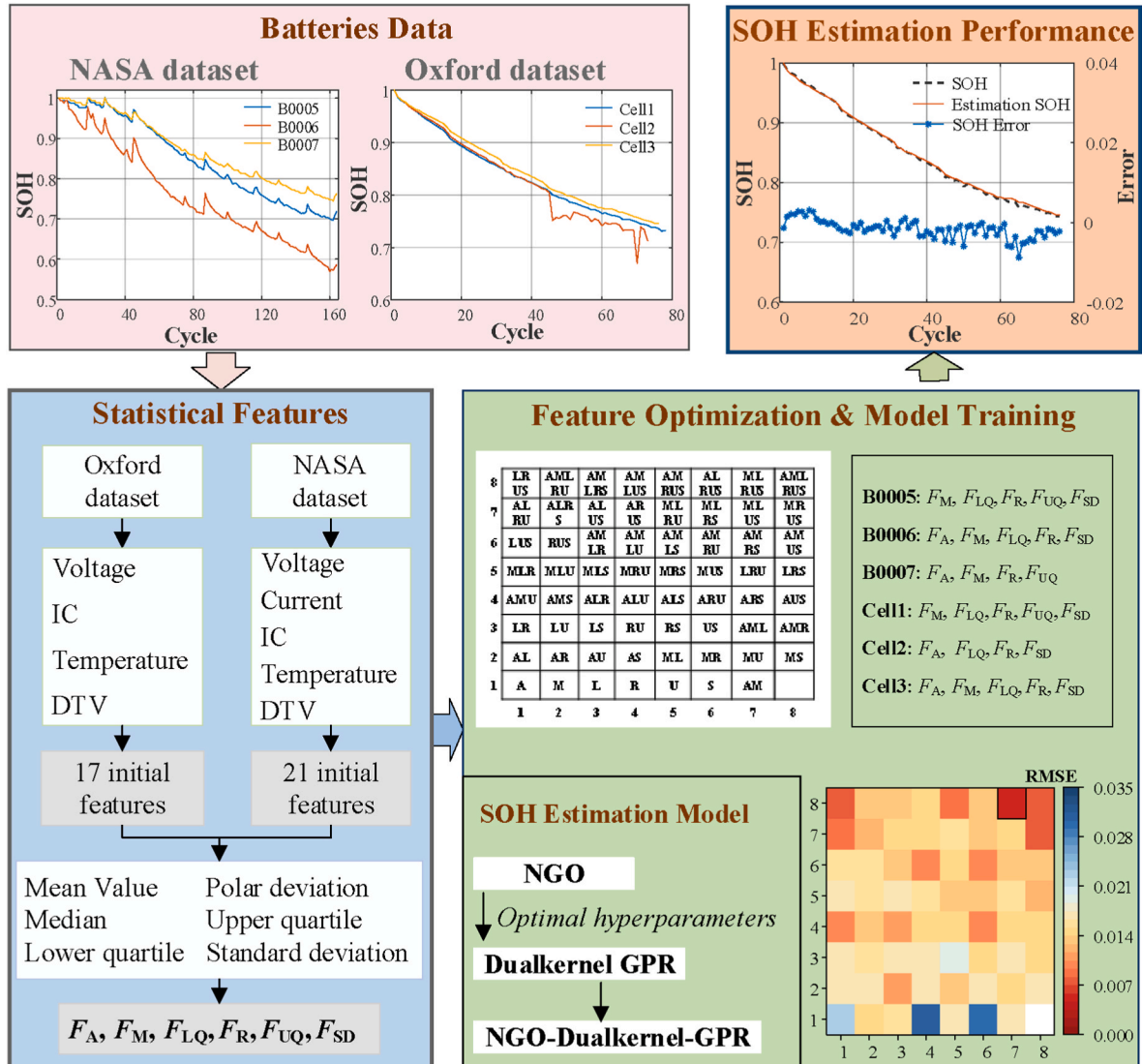


Fig. 8. Framework of feature optimization and SOH estimation.

Table 3
MAE (%) and RMSE (%) for the NASA dataset.

Cell		B0005	B0006	B0007
MAE	GPR	1.03%	0.80%	1.21%
	Dualkernel GRP	0.75%	0.53%	0.75%
	NGO-Dualkernel-GPR	0.59%	0.42%	0.60%
RMSE	GPR	1.35%	1.18%	1.40%
	Dualkernel GRP	0.99%	0.82%	0.95%
	NGO-Dualkernel-GPR	0.76%	0.64%	0.72%

combinatorial kernel function still has a significant estimation error on some loops. The Dualkernel GPR optimized by the NGO effectively reduces these errors. Compared with the previous two estimation methods, better estimation results are achieved. As shown in Table 3, the MAE and RMSE of single-kernel GPR on the NASA data sets are both within 1.5%, which verifies the validity of the statistical feature sets extracted in this study. The MAE and RMSE of Dualkernel GPR after optimization by NGO are significantly reduced compared with the other two methods. The RMSE and MAE for all three cells are within 1% and 0.6%, respectively.

4.1.2. Oxford dataset testing

Although the effectiveness of the proposed method has been verified

on the NASA dataset, the proposed method was also tested on the Oxford dataset to further prove its feasibility (i.e., generalization). Cell1, Cell2, and Cell3 were selected as the test datasets. All statistical features were extracted as model inputs for SOH estimation. Moreover, they are compared with the GPR model using a squared exponential covariance function as the kernel function, the Dualkernel GPR model with Ma3+Ma5, and the dual-kernel GPR model optimized by the NGO algorithm. The leave-one-out cross-validation method was employed for the test.

As shown in Fig. 10, the dual-kernel GPR optimized by NGO can better track the degradation process of the cell and obtain better SOH estimation accuracy. As the LIB ages gradually, the maximum error of the GPR is around 2%, confirming the effectiveness of the extracted statistical feature set in this study. In comparison, the maximum error of NGO-Dualkernel-GPR is controlled within 1%. Table 4 shows that compared to the GPR model, the NGO-Dualkernel-GPR model achieves a reduction of approximately 0.7% in RMSE on the three LIB cells. The Oxford data set experiments also confirm the effectiveness of the proposed method.

4.2. Statistical feature optimization for SOH estimation

In previous experiments, all statistical characteristics were adopted as inputs to estimate the battery SOH. However, the problems that could

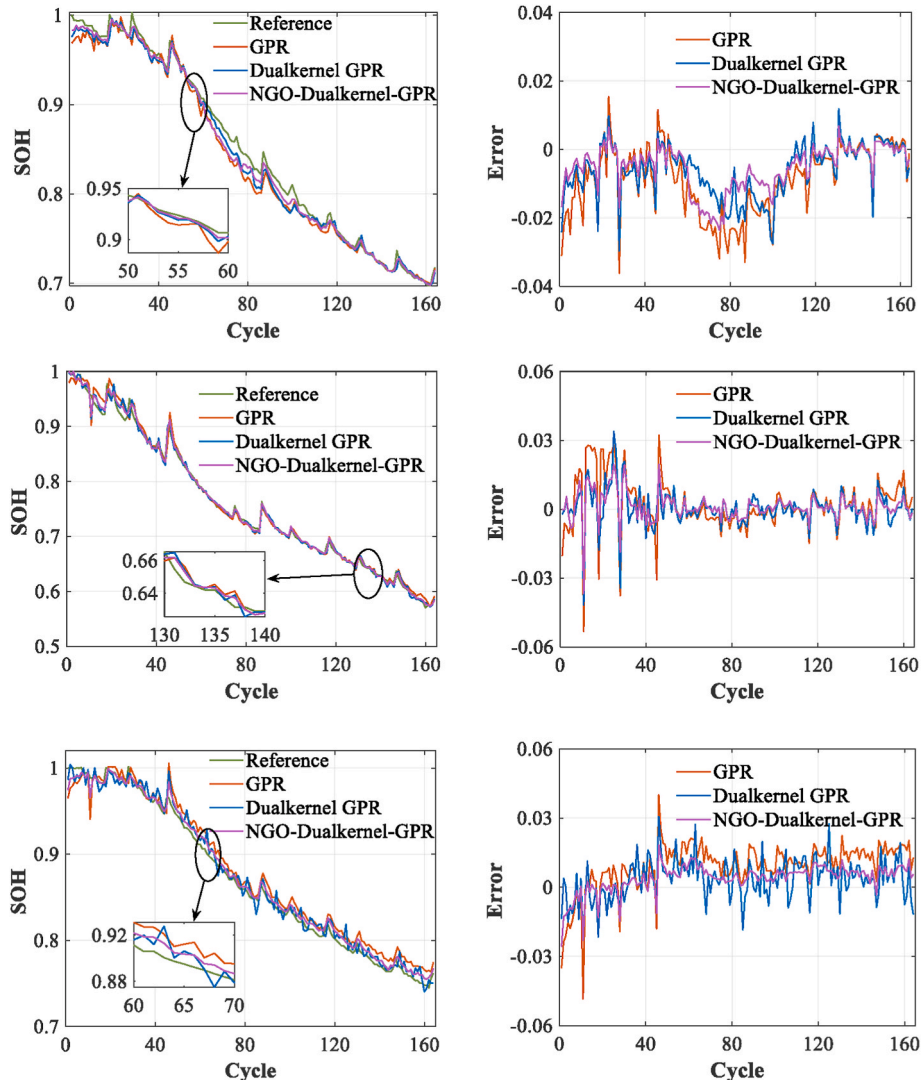


Fig. 9. Experimental results and error analysis of NASA data set.

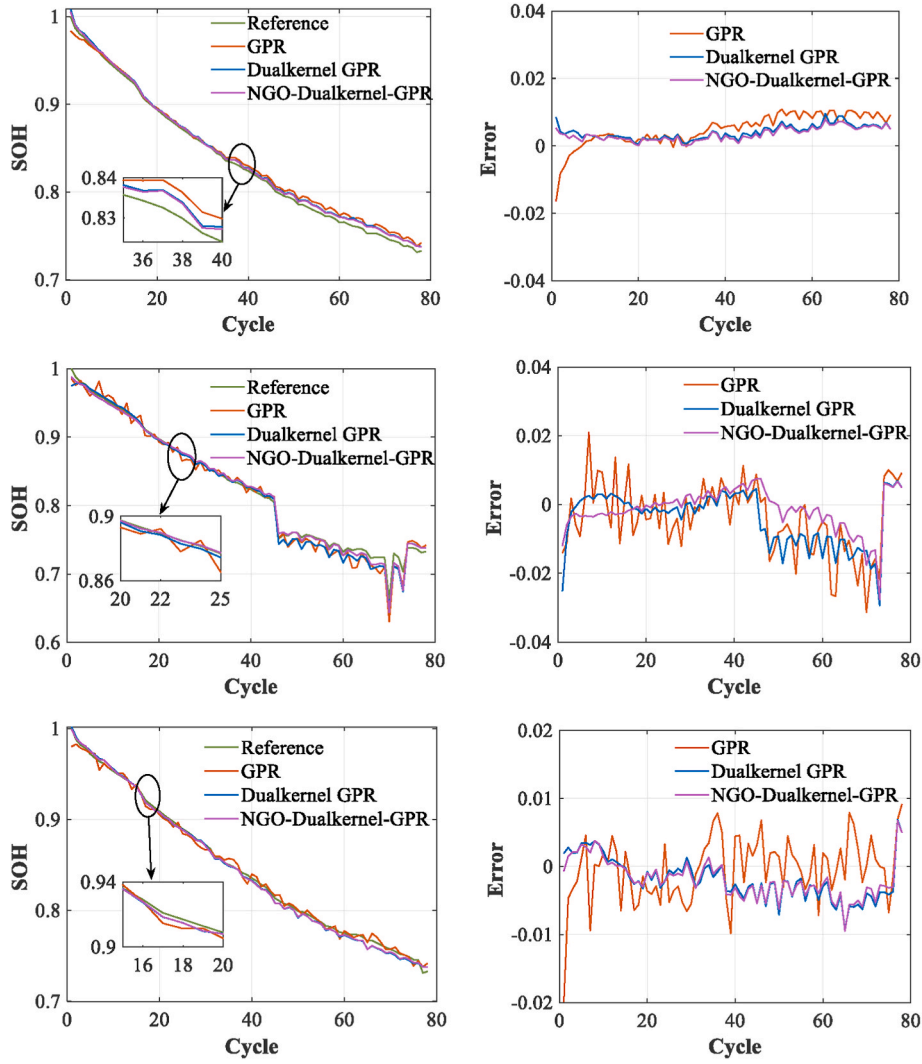


Fig. 10. Experimental results and error analysis of Oxford data set.

Table 4

MAE (%) and RMSE (%) for the Oxford dataset.

Cell		Cell1	Cell2	Cell3
MAE	GPR	0.58%	0.86%	0.36%
	Dualkernel GRP	0.41%	0.65%	0.29%
	NGO-Dualkernel-GPR	0.35%	0.42%	0.28%
RMSE	GPR	0.68%	1.11%	0.47%
	Dualkernel GRP	0.45%	0.91%	0.62%
	NGO-Dualkernel-GPR	0.40%	0.35%	0.33%

be caused by overfitting were ignored. Therefore, to assess the contribution of features to SOH more reasonably, this study evaluates all possible combinations of features to find the optimal combination of features for each cell by the method of permutation in mathematics.

4.2.1. NASA dataset experiments

To find the optimal combination of statistical feature sets for the three LIB groups of the NASA dataset. To this end, all possible permutations and combinations of the six statistical features are mathematically arranged and combined, as shown in Annex 1. Subsequently, this study acquires the minimum error of the combination of features. by training with a dual-kernel GPR after NGO optimization. As shown in Annex Tables 1 and 2, (i, j) indicates different statistical feature

combinations. Besides, the distribution of the specific statistical feature combinations is referred to Supplementary Table 2. In the feature combinations, B0005, B0006, and B0007 obtain the best cross-validation RMSE of 0.54%, 0.61%, and 0.6% at $(7, 8) = [F_M, F_{LQ}, F_R, F_{UQ}, F_{SD}]$, $(8, 8) = [F_A, F_M, F_{LQ}, F_R, F_{SD}]$ and $(6, 6) = [F_A, F_M, F_R, F_{UQ}]$, respectively.

As shown in Fig. 11, the combination of features after B0005 optimization for SOH estimation, the errors are within 2% for the entire cycle and within 1% for the vast majority of cycles. As shown in Table 2 in the Appendix, compared with all statistical feature sets, the MAE and RMSE of these three LIB cells decreased by (0.18%, 0.02%, 0.1%) and (0.22%, 0.03%, 0.12%), respectively. Experimental results demonstrate the effectiveness of the proposed statistical feature optimization method.

4.2.2. Oxford dataset experiments

To further verify the effectiveness and generalizability of the proposed method, experiments were conducted on the Oxford dataset with three different types of cells (Cell1, Cell2, Cell3). (i, j) denotes different statistical feature combinations, where the distribution of the array of specific statistical feature combinations refers to Supplementary Table 2. In the feature combination, Cell1, Cell2, and Cell3 obtained the best cross-validation RMSE of 0.32%, 0.37%, and 0.27% at $(6, 7) = [F_M, F_{LQ}, F_R, F_{UQ}, F_{SD}]$, $(2, 7) = [F_A, F_{LQ}, F_R, F_{SD}]$ and $(3, 8) = [F_A, F_M, F_{LQ}, F_R, F_{SD}]$ respectively.

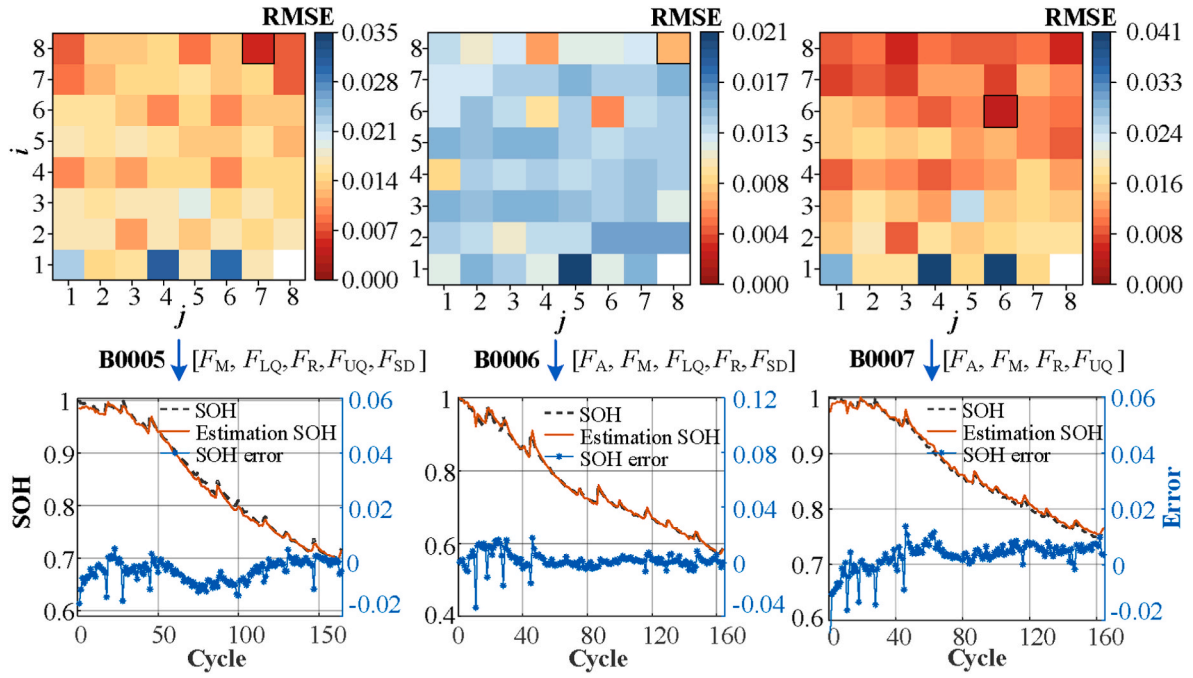


Fig. 11. RMSE comparison plots of the cross-validation set of 64 statistical feature combinations based on the NASA dataset (B005, B006, B007) (top panel) and the SOH estimation performance of the optimal statistical feature combination (bottom panel).

As shown in Fig. 12, the feature set after feature optimization can contain as much information as possible to reflect the degradation of the cells. The estimated battery SOH can well reflect the degradation trend. The errors of Cell1 and Cell3 in the three cells are within 1%, while most of the errors in Cell2 are within 1%, and only very few cycles have errors more significant than 1%, but they are all within 2%. As shown in Table 2 in the Appendix, the RMSE and MAE of the optimized feature combinations on the three Cells are within 0.4% and 0.3%, respectively. The estimation accuracy is also significantly improved compared to the unoptimized statistical feature set. Experimental results on the Oxford

dataset also confirm the effectiveness of the proposed feature optimization method.

4.3. Experimental discussion

Considering the different operating conditions of NASA and Oxford datasets, the experimental extract 21 and 16 initial charging features, such as voltage, current, IC, and DT, from NASA and Oxford datasets, respectively. To minimize the impact of sensor noise and redundant features, an optimal statistical feature set of each charge cycle for SOH

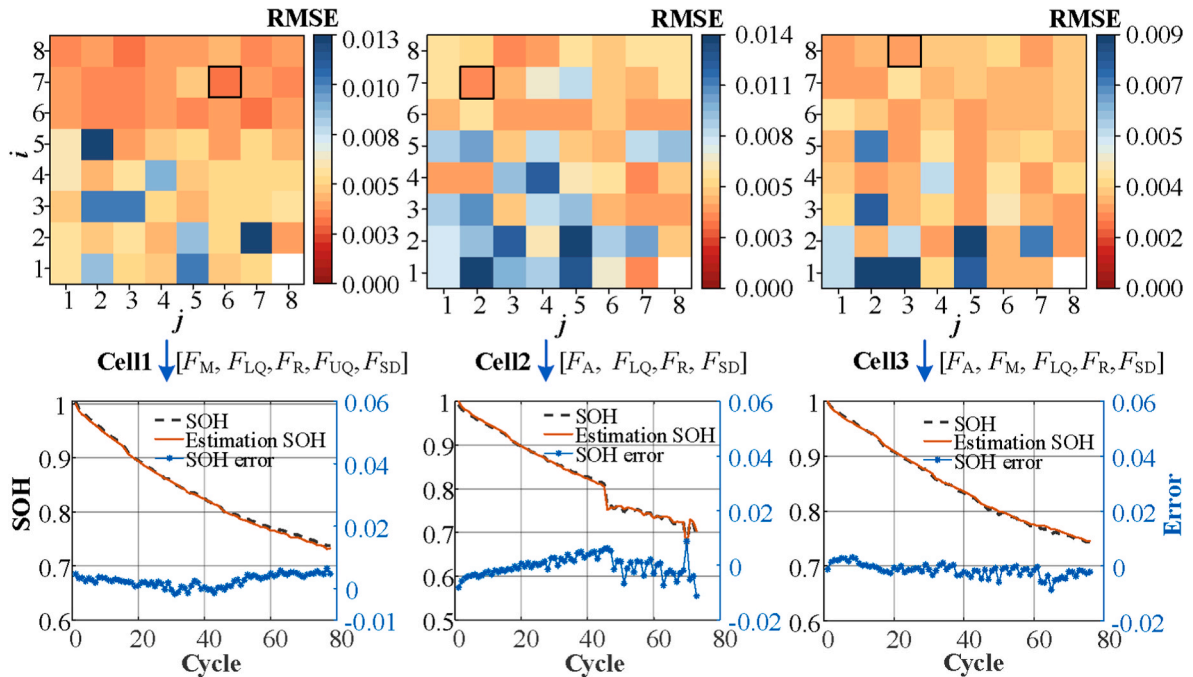


Fig. 12. RMSE comparison plots of the cross-validation set of 64 statistical feature combinations based on the Oxford dataset (Cell1, Cell2, Cell3) (top panel) and the SOH estimation performance of the optimal statistical feature combination (bottom panel).

estimation was acquired by a comparative analysis among different combinations of the 6 statistical features (i.e., mean, median, lower quartile, range, upper quartile, and standard deviation).

The experimental results indicate that the proposed method NGO-Duakernel-GPR has higher accuracy and better robustness. Compared with GPR and Duakernel-GPR methods, the maximum error of the NGO-Duakernel-GPR method is less than 1%. In the NASA dataset, the MAE values of B005, B006, and B007 are 0.59%, 0.42%, and 0.60%, respectively, the RMSE values of B005, B006, and B007 are 0.76%, 0.64%, and 0.72%, respectively. In the Oxford dataset, the MAE values of Cell1, Cell2, and Cell3 are 0.35%, 0.42%, and 0.28%, respectively, the RMSE values of Cell1, Cell2, and Cell3 are 0.40%, 0.35%, and 0.33%, respectively.

5. Conclusion

This study proposes a statistical feature optimization method based on the dual-kernel GPR with NGO optimization to improve the SOH estimation accuracy, we employed a Duakernel GPR model. Instead of the manual calculation, the hyperparameters of the Duakernel GPR model were optimized by the NGO algorithm. Compared with standard GPR, the dual-kernel function can effectively track both the long-term downward trend and short-term variation in LIB capacity. However, the Duakernel GPR model does not guarantee whether the value of the obtained hyperparameter set is optimal. This study adopted NGO for hyperparameter optimization. This proposed algorithm can automatically search for the optimal set of hyperparameters, thereby improving the accuracy of modeling prediction.

The proposed method can estimate battery SOH with high accuracy

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2023.119907>.

Appendix A

The 8×8 array for 64 combinations of 6 statistical features (A: F_A , M: F_M , L: F_{LQ} , R: F_R , U: F_{UQ} , S: F_{SD}).

Table 1

$i(F)$	8	LRUS	AMLRU	AMLRS	AMLUS	AMRUS	ALRUS	MLRUS	AMLRUS
	7	ALRU	ALRS	ALUS	ARUS	MLRU	MLRS	MLUS	MRUS
	6	LUS	RUS	AMLR	AMLU	AMLS	AMRU	AMRS	AMUS
	5	MLR	MLU	MLS	MRU	MRS	MUS	LRU	LRS
	4	AMU	AMS	ALR	ALU	ALS	ARU	ARS	AUS
	3	LR	LU	LS	RU	RS	US	AML	AMR
	2	AL	AR	AU	AS	ML	MR	MU	MS
	1	A	M	L	R	U	S	AM	
		1	2	3	4	5	6	7	8
	$j(F)$								

References

[1] K. Liu, B. Lin, Research on influencing factors of environmental pollution in China: a spatial econometric analysis, *J. Clean. Prod.* 206 (2019) 356–364.

[2] X. Huang, J. Ge, Electric vehicle development in Beijing: an analysis of consumer purchase intention, *J. Clean. Prod.* 216 (2019) 361–372.

[3] W. Zeng, F. Xia, W. Tian, et al., Single-crystal high-nickel layered cathodes for lithium-ion batteries: advantages, mechanism, challenges and, *Curr. Opin. Electrochem.* 31 (2022) 100831.

[4] X. Zhang, Z. Li, L. Luo, et al., A review on thermal management of lithium-ion batteries for electric vehicles, *Energy* 238 (2022) 121652.

[5] X. Li, Z. Wang, L. Zhang, Co-estimation of capacity and state-of-charge for lithium-ion batteries in electric vehicles, *Energy* 174 (2019) 33–44.

[6] L. Zhang, C. Zhu, S. Yu, et al., Status and challenges facing representative anode materials for rechargeable lithium batteries, *J. Energy Chem.* 66 (2022) 260–294.

[7] X. Liu, X.Q. Zhang, X. Chen, et al., A generalizable, data-driven online approach to forecast capacity degradation trajectory of lithium batteries, *J. Energy Chem.* 68 (2022) 548–555.

[8] H. Tian, P. Qin, K. Li, et al., A review of the state of health for lithium-ion batteries: research status and suggestions, *J. Clean. Prod.* 261 (2020) 120813.

[9] X. Hu, F. Feng, K. Liu, et al., State estimation for advanced battery management: key challenges and future trends, *Renew. Sustain. Energy Rev.* 114 (2019) 109334.

[10] US ABC electric vehicle battery test procedures manual, revision 2. Principal author: gary hunt, Idaho National Engineering Laboratory (INEL), *US Department of energy Idaho Field Office*, DOE/ID-10479, Review 2 (1996) 39.

[11] P. Tagade, K.S. Hariharan, S. Ramachandran, et al., Deep Gaussian process regression for lithium-ion battery health prognosis and degradation mode diagnosis, *J. Power Sources* 445 (2020) 227281.

[12] X. Tang, K. Liu, J. Lu, et al., Battery incremental capacity curve extraction by a two-dimensional Luenberger-Gaussian-moving-average filter, *Appl. Energy* 280 (2020) 115895.

and stability for different LIB datasets with different operating conditions, which means the proposed feature optimization method analyzes the health features in the LIB charging process through statistical methods, greatly simplifying the computational effort involved in SOH estimation while maintaining accuracy. The proposed lightweight SOH estimation method can provide adequate support for the safe and reliable operation of LIBs.

Thus, this study has a broad application prospect and can provide critical support for the health management and safety control of LIBs.

Funding

Fujian Provincial Science and Technology Program 2023T3084 and 2023T3069.

Credit author statement

Houde Dai, and Jiaxin Wang: Software, Writing e original draft preparation, Formal analysis, Data curation, Validation.

Liqi Zhu: Conceptualization, Methodology, Investigation, Writing Reviewing and Editing, Supervision.

Liqi Zhu: Resources, Data curation, Visualization, Validation.

Yiyang Huang, and Yuan Lai: Reviewing and Editing.

Declaration of competing interest

The author has no relevant financial or non-financial interests to disclose. There is no conflict of interest to publish this work in this journal.

- [13] H. Sun, J. Sun, K. Zhao, et al., Data-driven ICA-Bi-LSTM-combined lithium battery SOH estimation, *Math. Probl. Eng.* 2022 (2022) 1–8.
- [14] Z. Deng, X. Hu, P. Li, et al., Data-driven battery state of health estimation based on random partial charging data, *IEEE Trans. Power Electron.* 37 (5) (2021) 5021–5031.
- [15] Z. Xu, J. Wang, P.D. Lund, et al., Co-estimating the state of charge and health of lithium batteries through combining a minimalist electrochemical model, *Energy* 240 (2022) 122815.
- [16] M. Ahwiadi, W. Wang, An enhanced mutated particle filter technique for system state estimation and battery life prediction, *IEEE Trans. Instrum. Meas.* 68 (3) (2018) 923–935.
- [17] B. Jiang, J. Zhu, X. Wang, et al., A comparative study of different features extracted from electrochemical impedance spectroscopy in state of health estimation for lithium-ion batteries, *Appl. Energy* 322 (2022) 119502.
- [18] C. Lyu, Q. Lai, T. Ge, et al., A lead-acid battery's remaining useful life prediction by using electrochemical model in the Particle Filtering framework, *Energy* 120 (2017) 975–984.
- [19] J. Li, L. Wang, C. Lyu, et al., New method for parameter estimation of an electrochemicalthermal coupling model for LiCoO₂ battery, *J. Power Sources* 307 (2016) 220–230.
- [20] M. Yu, Y. Li, I. Podlubny, et al., Fractional-order modeling of lithium-ion batteries using additive noise assisted modeling and correlative information criterion, *J. Adv. Res.* 25 (2020) 49–56.
- [21] Q. Zhang, C.G. Huang, H. Li, et al., Electrochemical impedance spectroscopy based state-of-health estimation for lithium-ion battery considering temperature and state-of-charge effect, *IEEE Trans. Transport. Electr.* 8 (4) (2022) 4633–4645.
- [22] S. Hosseinasab, C. Lin, S. Pischinger, et al., State-of-health estimation of lithium-ion batteries for electrified vehicles using a reduced-order electrochemical model, *J. Energy Storage* 52 (2022) 104684.
- [23] J. Deng, C. Bae, A. Denlinger, et al., Electric vehicles batteries, *Joule* 4 (3) (2020) 511–515.
- [24] S. Wang, D.I. Stroe, C. Fernandez, et al., A novel energy management strategy for the ternary lithium batteries based on the dynamic equivalent circuit modeling and differential Kalman filtering under time-varying conditions, *J. Power Sources* 450 (2020) 227652.
- [25] S. Khaleghi, M.S. Hosen, D. Karimi, et al., Developing an online data-driven approach for prognostics and health management of lithium-ion batteries, *Appl. Energy* 308 (2022) 118348.
- [26] P. Li, Z. Zhang, R. Grosu, et al., An end-to-end neural network framework for state-of-health estimation and remaining useful life prediction of electric vehicle lithium batteries, *Renew. Sustain. Energy Rev.* 156 (2022) 111843.
- [27] A. Manoharan, K.M. Begam, V.R. Aparow, et al., Artificial neural networks, gradient boosting and support vector machines for electric vehicle battery state estimation: a review, *J. Energy Storage* 55 (2022) 105384.
- [28] Y. Li, M. Abdel-Monem, R. Gopalakrishnan, et al., A quick on-line state of health estimation method for Li-ion battery with incremental capacity curves processed by Gaussian filter, *J. Power Sources* 373 (2018) 40–53.
- [29] S.S.Y. Ng, Y. Xing, K.L. Tsui, A naive Bayes model for robust remaining useful life prediction of lithium-ion battery, *Appl. Energy* 118 (2014) 114–123.
- [30] Y. Zhang, Z. Peng, Y. Guan, et al., Prognostics of battery cycle life in the early-cycle stage based on hybrid model, *Energy* 221 (2021) 119901.
- [31] F. Xu, F. Yang, Z. Fei, et al., Life prediction of lithium-ion batteries based on stacked denoising autoencoders, *Reliab. Eng. Syst. Saf.* 208 (2021) 107396.
- [32] Y. Ma, C. Shan, J. Gao, et al., A novel method for state of health estimation of lithium-ion batteries based on improved LSTM and health indicators extraction, *Energy* 251 (2022) 123973.
- [33] J. Wen, X. Chen, X. Li, et al., SOH prediction of lithium battery based on IC curve feature and BP neural network, *Energy* 261 (2022) 125234.
- [34] J. Wang, R. Zhao, Q.A. Huang, et al., High-efficiency prediction of state of health for lithium-ion battery based on AC impedance feature tuned with Gaussian process regression, *J. Power Sources* 561 (2023) 232737.
- [35] H.H. Goh, Z. Lan, D. Zhang, et al., Estimation of the state of health (SOH) of batteries using discrete curvature feature extraction, *J. Energy Storage* 50 (2022) 104646.
- [36] S. Son, S. Jeong, E. Kwak, et al., Integrated framework for SOH estimation of lithium-ion batteries using multiphysics features, *Energy* 238 (2022) 121712.
- [37] M. Lin, D. Wu, J. Meng, et al., Health prognosis for lithium-ion battery with multi-feature optimization, *Energy* 264 (2023) 126307.
- [38] B. Saha, K. Goebel, *Battery Data Set, NASA Ames Prognostics Data Repository, Moffett Field, CA.* <http://ti.arc.nasa.gov/tech/dash/pcoe/prognosticdatarepository/>, 2007.
- [39] C. Birkl, *Diagnosis and Prognosis of Degradation in Lithium-Ion batteries*[D], University of Oxford, 2017.
- [40] D. Zhou, W. Zheng, S. Chen, et al., Research on state of health prediction model for lithium batteries based on actual diverse data, *Energy* 230 (2021) 120851, 0.
- [41] P. Sedgwick, Pearson's correlation coefficient, *BMJ* (2012) 345.
- [42] M. Dehghani, Š. Hubálovský, P. Trojovský, Northern goshawk optimization: a new swarm-based algorithm for solving optimization problems, *IEEE Access* 9 (2021) 162059–162080.
- [43] E. Schulz, M. Speekenbrink, A. Krause, A tutorial on Gaussian process regression: modelling, exploring, and exploiting functions, *J. Math. Psychol.* 85 (2018) 1–16.
- [44] M. Cheng, X. Zhang, A. Ran, et al., Optimal dispatch approach for second-life batteries considering degradation with online SoH estimation, *Renew. Sustain. Energy Rev.* 173 (2023) 113053.